

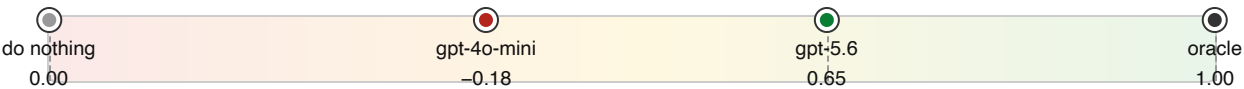
er-gym — AI benchmark results

Can a language model run an emergency department shift?

Scenario: **ed-baseline** · seed **s1** · one 12-hour shift (144 five-minute rounds) · single seed

The scale

Each model plays the same shift and is scored against two fixed reference runs on the identical seed: **doing nothing** (the floor, 0.00) and a **perfect-information oracle** (the ceiling, 1.00). A model at 0.5 closed half the gap between abandoning the department and flawless play. **Below 0 is worse than doing nothing** — possible here because the hard safety penalties only fire when you *act*.



Marks placed on the 0–1 scale (–0.18 sits left of the "do nothing" line).

Results

Model	Score	Deaths	Walked out	Roomed	Sent onward	Verdict
gpt-5.6 (reasoning)	0.65	2	27%	48	16	Runs the department
gpt-4o-mini	–0.18	10	71%	0	0	Froze — worse than nothing
gpt-3.5-turbo	no data	—	—	—	—	Run not completed (budget)

"Roomed" = patients placed in a bed so a doctor could see them. "Sent onward" = discharged or admitted out. Both are zero for a model that never advances care.

What actually happened

The benchmark discriminates cleanly, and it caught two distinct failure modes.

- **gpt-5.6** played it like a charge nurse: it triaged, put 48 patients in beds, ordered workups, sent 16 onward, and triaged the ringing phones (deferring the noise). Two patients died, 27% left without being seen — far from perfect, but it genuinely ran the floor.
- **gpt-4o-mini** never formed the plan. It measured vitals **~1,300 times** and never once put a patient in a bed, assigned a doctor, or ordered a treatment. Patients piled up in the waiting room; 71% walked out, 10 died — the same outcome as an empty department, but scored slightly worse because it broke a few paperwork rules while achieving nothing.

The gap is not about the interface. gpt-4o-mini was handed the list of legal moves — including "place patient in a bed" — on every single round, and still never did it. That rules out "it didn't know it could" and points at a planning ceiling: the small model fixates on the single cheapest legal action and cannot sequence *room* → *treat* → *send onward*.

Reading the safety trade-off

gpt-5.6's higher score came partly from working aggressively, and that cost it on safety: it committed **30 safety-floor violations** (mostly nurse-ratio breaches from overloading staff, plus one medication-verification bypass and one controlled-substance slip). gpt-4o-mini's only violations were missed reporting clocks. In other words, the model that actually treated patients also took on the real risk that comes with acting — which is exactly the tension the benchmark is built to measure.

Limits of this run

- **Single seed.** One shift each. The exact scores carry roughly ± 0.2 of noise; treat 0.65 and -0.18 as "clearly above" and "clearly below" doing nothing, not precise figures. Multiple seeds are needed to quote a number.
- **gpt-3.5-turbo has no result** — its run was cut off when the API budget ran out. Not reported rather than reported wrong.
- **Scored with a straightforward prompt.** A stronger scaffold could lift any model; the number reflects model + prompt together.
- **Simulator parameters are plausible, not fitted** to real emergency-department data.

Bottom line

The benchmark works: it separates a model that can run an ED (gpt-5.6, 0.65) from one that cannot (gpt-4o-mini, below zero), and it explains *why* in terms of what each did on the floor rather than a single opaque score. The next step is multiple seeds and more models to turn these single-shift readings into stable numbers.